

# Body and Office Chairs---The Science of Sitting

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## Abstract

*Man's body is not straight, which is determined by biomorph. Therefore, a "competent" office chair must be consistent with the shape of human body, and their characteristics of movement, etc. From the perspective of ergonomics, this thesis aims to talk about some characteristics that an excellent office chair should have.*

**Keywords:** ergonomics, pressure distribution, chair design

People can not live without chairs in their work and life, especially those people who need to spend more than 1 / 3 of their daily time sitting in chairs. Thus besides the properly used materials and generous beautiful shape, the design of office chairs has to meet the principles of ergonomic design, that is to say, a seat should be designed to fully consider the body's physiological characteristics of sitting state, with its aim to put the sitting people in a better state.

On the whole, based on ergonomics, several factors need to be taken into account in design of offices, including the anthropometrics of fit, body support, pressure distribution, the kinematics of sitting, stabilization, and thermal comfort.

## 1. The Anthropometrics of Fit

A chair should fit the body like a piece of clothing. Ideally, a work chair should provide the personal fit of a shirt or a pair of pants. People shouldn't be required to "wear" chairs that are too big or too small.

People vary considerably in shape as well as overall size. Even among a group of people of the same gender, age, and stature, one finds significant variation in bodily proportions. Two men of the same standing height, for instance, can appear to be of very different heights when seated, and their seated elbow heights may vary by as much as three or four inches. Achieving a match between certain body dimensions and corresponding chair dimensions is crucial to the sitter's comfort and

health. The wrong seat height can cause uncomfortable pressure on the backs of the thighs<sup>[1]</sup>. A seat pan that is too wide or too deep may prevent the sitter from taking advantage of armrests and backrest contours that help to transfer weight from the spine.

So, we must gather a lot of anthropometric data before design a work chair. And to gather our own anthropometric data, we can take seven important measurements:

- popliteal height (lower leg length)
- seat depth (buttock to popliteal length)
- hip breadth
- midshoulder sitting height (back height)
- elbow height
- lumbar height
- lumbar depth

## 2. Body Support

Good seating design can improve anatomical performance by lessening ergonomic stresses on the body. Although many of these stresses can be reduced by changing sitting behaviors, seating designers attempt to alleviate them through the design of the chair. This is not a simple task; the interaction between body, posture, the work surface, personal preference, and the chair can be complex. Can chair design, or chair comfort, increase productivity? A survey of office workers found that employees believe that increased comfort would result in increased productivity, and experts often agree<sup>[2]</sup>. More than one study has documented significant performance improvements (as much as 24 percent) among clerical workers moved to appropriately sized and comfortable furniture, including chairs. Chairs can have an impact on back comfort and health to the extent that they affect the major ergonomic risk factors of sitting:

- sustained static postures
- direct loads on the spine
- loads on the spine that are magnified by flattened lower backs and other postures
- pressure on the skin and underlying tissues
- increased musculoskeletal stress

The risk factors can be translated into design objectives for office seating.

1. Allow or encourage varied posture and posture change.
2. Reduce muscle activity required while sitting.
3. Reduce the weight carried by the lower spine.
4. Encourage natural lower back shape (lordosis).
5. Reduce contact pressure on the skin and other tissues.
6. Fit different body sizes.

### 3. Pressure Distribution

The perfect work chair would conform equally well to all body shapes, sizes, and contours without applying circulation-restricting pressure anywhere. Surface pressure can cause discomfort while sitting. People of different body weights and builds distribute their weight on a chair in similar patterns, but pressure intensity and areas of distribution vary from person to person<sup>[3]</sup>. Good pressure distribution in a chair focuses peak pressure under the sitting bones in upright postures and in the lumbar and thoracic areas in reclined postures.

Figure 1

Sitting in a reclined position in a chair with topographically neutral support distributes pressure across the thoracic area and away from the spine.

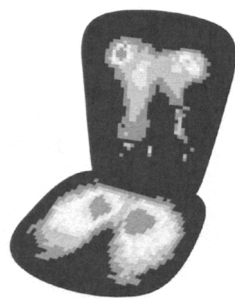


Figure.1

A comfortable chair will produce pressure distributions for a wide anthropometric range of users that show peaks in the area of the ischial tuberosities when the sitter is in an upright posture and areas of the back away from the spine when the sitter is in a reclining posture (Figure 1).

Design a chair that is “topographically neutral,” so that the sitter’s body, and not the underlying structures of the seat pan and backrest, determines peak pressure areas. Minimize chair structure and eliminate the need for foam padding by tensioning a material that provides dynamic support. Engineer the chair so the structure fits differently proportioned persons.

### 4. The Kinematics of Sitting

A chair should move the way the body moves. In the best of all possible worlds, the body is free to position itself spontaneously, constrained only by gravity. A person seated at work should be able to move freely and unselfconsciously from computer-related tasks to more relaxed or interactive postures. The work chair should follow along, providing optimal support whether the body is in motion or at rest.

The seated body has natural pivot points at the hip, knee, and ankle. The ideal work chair will mimic as closely as possible those pivot points, thus allowing the sitter to move naturally and with ease around the chair’s engineered pivot points<sup>[4]</sup>. The pivot points within the user’s body and the movement of the chair should work together to create a tilt movement that doesn’t stress or lift the body. The more closely a work chair moves like the body, the more likely it is that the seated user will move his or her body.

The linkages of the chair were designed to echo these body linkages in a movement that corresponds to the natural movement of the human body. We call it the kinematic coherence model. It would look like (Figure 2):

- With the feet flat on the floor and a slightly open angle between the lower and upper leg, all of the body’s major joints pivot about the center of the ankle joint (A).

Figure 2

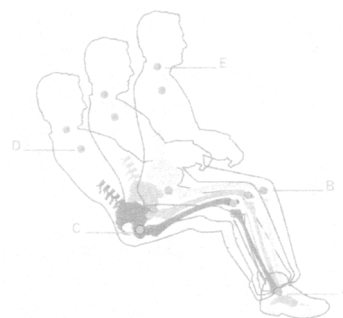


Figure.2

- The knee joint (B) actually doesn’t pivot so much as travel in a gentle arc.
- The hip joint (C) follows an inclined arched path about a line connected to the ankle pivot (A) and pivots as the trunk-to-thigh angle opens.
- The arms pivot about the shoulder joint (D).
- Throughout the movement the head pivots about the cervical spine (E) to maintain a constant relationship between the face plane and a visual target.

## 5. Stabilization

As we know: People are more likely to sit in upright or forward-leaning postures than to recline while working at the computer. Lumbar support alone cannot naturally restore spinal curvatures in a body that is seated in an upright position. The pelvis serves as the body's center of mass and the primary attachment location for muscles that move the upper and lower body. When the pelvis rotates in a rearward position, the body works to restore balance at the cost of muscle fatigue and discomfort.

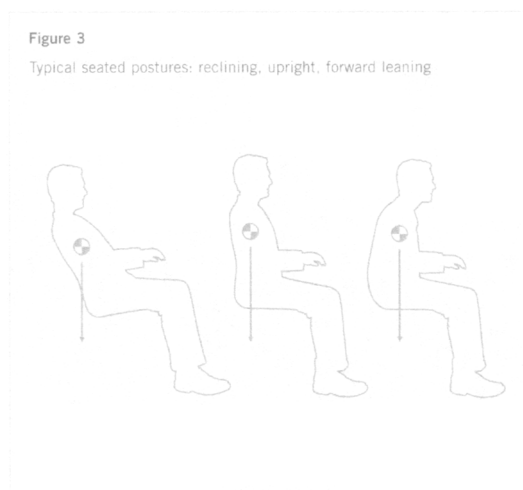


Figure.3

For the purposes of studying the seated human body at work, ergonomists have identified three postures based on the location of the body's center of mass: reclining, upright, and forward leaning (Figure 3). A good work chair should preserve natural spinal alignment, even in upright postures. Reclining postures help to maintain the natural curve of the spine by opening the angle between torso and thigh to attain pelvic alignment. The same effect can be achieved in upright postures by stabilizing the sacral-pelvic area of the back to sustain the forward pelvic tilt that promotes natural spinal curvatures and muscle balance.

## 6. Thermal Comfort

People are more comfortable when ambient temperatures are neutral, allowing the body to maintain thermal equilibrium without sweating or shivering. Increased humidity at the skin's surface can lead to sitting discomfort.

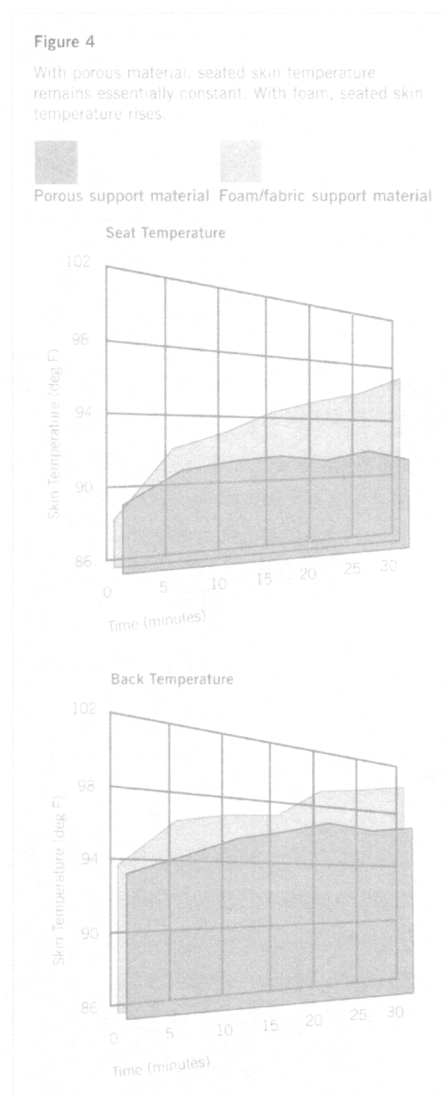


Figure.4

A chair should breathe, it should have a neutral effect on body-surface temperatures, so that thermal comfort is not posture dependent, and it will have a neutral effect on thermal comfort by allowing the flow of heat or the dispersion of water vapor generated by the body. So, with porous material, seated skin temperature remains essentially constant. With foam, seated skin temperature rises.

The sitting temperatures remained essentially constant and comparable to their standing temperatures even after 30 minutes sitting in chairs with breathable materials. After 30 minutes seated in the foam chairs, however, the temperatures were higher by about 10.7 degrees F (Figure 4).

End

To sum up, the design of office chairs is a very complex, so designers need to consider the above factors first when they design chairs. Comfortable chairs with beautiful shape make people enjoy sitting and applaud its comfort and convenience.

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